

# Manual Execution for Problem 4

## Graph with 3 connected components

Source file: 'data/manual/undirectedComponents1.txt'

Vertices: '0 1 2 3 4 5 6 7'

Edges:

- 0 -- 1

- 1 -- 2

- 2 -- 3

- 3 -- 0

- 0 -- 2

- 4 -- 5

- 5 -- 6

- 4 -- 6

Vertex '7' is isolated.

### BFS connected components

Traversal:

| Step | Queue before pop | Popped | Visited after step |
|------|------------------|--------|--------------------|
| 1    | 0                | 0      | {0, 1, 2, 3}       |
| 2    | 1, 2, 3          | 1      | {0, 1, 2, 3}       |
| 3    | 2, 3             | 2      | {0, 1, 2, 3}       |
| 4    | 3                | 3      | {0, 1, 2, 3}       |

First component: '{0, 1, 2, 3}'

Restart from '4':

| Step | Queue before pop | Popped | Visited after step    |
|------|------------------|--------|-----------------------|
| 5    | 4                | 4      | {0, 1, 2, 3, 4, 5, 6} |
| 6    | 5, 6             | 5      | {0, 1, 2, 3, 4, 5, 6} |
| 7    | 6                | 6      | {0, 1, 2, 3, 4, 5, 6} |

Second component: '{4, 5, 6}'

Restart from '7':

| Step | Queue before pop | Popped | Visited after step       |
|------|------------------|--------|--------------------------|
| 8    | 7                | 7      | {0, 1, 2, 3, 4, 5, 6, 7} |

Third component: '{7}'

## Graph with 1 connected component

Source file: 'data/manual/undirectedComponents2.txt'

Edges:

- 0 -- 1
- 1 -- 2
- 2 -- 3
- 3 -- 4
- 4 -- 5
- 5 -- 0
- 0 -- 2
- 1 -- 4

### BFS connected components

Starting from '0', the queue expands in this order:

'0 -> 1 -> 2 -> 5 -> 3 -> 4'

Connected components: '{0, 1, 2, 3, 4, 5}'